

Programming Assignment Report

Design and Analysis of Algorithms

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Problem Name:	Maximum Subarray Sum
Problem HackerRank Link:	https://www.hackerrank.com/challenges/maximum-subarray-sum/problem
Github Submission Link:	https://github.com/harshithsuresh696-arch/daa_assignment/tree/main

1. Problem Explanation

We are given an array of integers and a number m . The goal is to determine the maximum value of the sum of any subarray modulo m . A subarray is a contiguous part of the array. Because the array size can be large (up to 10^5), checking all subarrays directly would be inefficient. Therefore, an optimized algorithm using prefix sums and TreeSet is used.

2. Example

Subarray	Sum	Sum % 7
[3]	3	3
[3,3]	6	6
[3,3,9]	15	1
[3,3,9,9]	24	3
[3,3,9,9,5]	29	1

Maximum value = 6

3. Approach / Algorithm

1. Compute prefix sum modulo m . 2. Store prefix sums in a TreeSet. 3. For each prefix value find the smallest prefix greater than the current prefix. 4. Compute candidate value using $(\text{prefix} - \text{next} + m) \% m$. 5. Update the maximum result and continue.

4. Time Complexity

Prefix computation takes $O(n)$. TreeSet operations take $O(\log n)$. Therefore overall complexity is $O(n \log n)$.

Space Complexity: $O(n)$

5. Solution

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

✔ Sample Test case 1

✔ Sample Test case 2

Input (stdin)[Download](#)

1	1
2	5 7
3	3 3 9 9 5

Your Output (stdout)

1	6
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Expected Output[Download](#)

1	6
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6. Running Testcases

✔ Test case 0

✔ Test case 4

✔ Test case 8

✔ Test case 12

✔ Test case 16

✔ Test case 1

✔ Test case 5

✔ Test case 9

✔ Test case 13

✔ Test case 17

✔ Test case 2

✔ Test case 6

✔ Test case 10

✔ Test case 14

✔ Test case 18

✔ Test case 3

✔ Test case 7

✔ Test case 11

✔ Test case 15

7. Observation

• Works correctly for the given sample input and output. • Efficient for large constraints. • All HackerRank test cases passed successfully.